Proposal for

Deep Learning PROJECT
Umm Al Qura University

Project Title: **Real-Time Face Mask Detection Using Deep Learning**

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Introduction

Face mask detection is becoming more important, particularly in safety and health monitoring systems. In situations where safety regulations must be followed, the ability to automatically detect whether someone is wearing a mask is critical.

Traditional methods rely on manual monitoring or rule-based image processing, which are often inefficient. Recent advances in deep learning have enabled models to learn complex facial representations, leading to more accurate and reliable detection.

Problem statement

Face mask detection remains challenging due to variations in lighting, face orientation, occlusion, and mask appearance. Conventional methods struggle to effectively handle such variability.

This project uses deep learning techniques to automatically classify masked and unmasked faces. By leveraging transfer learning and convolutional neural networks, the proposed system seeks to develop a robust and accurate face mask detection model.

Deep learning is particularly suitable for this task due to its ability to extract high-level visual features while also handling complex real-world variations in images.

Objectives

1.Implement a deep learning model to detect whether a person is wearing a mask or not.

2.Improve the model by training it using images of people wearing masks and people not wearing masks.

3.Evaluate the model performance by analyzing the confusion matrix, training and validation accuracy/loss curves, and F1-score.

4.Analyze and interpret the results to assess the model’s effectiveness in detecting face masks.

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Dataset and Data Preparation

The selected dataset for this project is the Face Mask 12K Images Dataset obtained from Kaggle (Jangra, 2020). PipelineThis dataset contains approximately 12,000 labeled facial images categorized into two classes: individuals wearing face masks and individuals not wearing face masks. The dataset is organized into separate training, validation, and testing subsets with a balanced distribution between the two classes.

This dataset was chosen due to its adequate size, structured organization, and suitability for binary image classification tasks. The images represent real-world variations in lighting conditions, facial orientations, background complexity, and mask types, which enhances the model's ability to generalize effectively to unseen data.

Planned methodology/architecture

Methodology This project adopts a deep learning approach for face mask classification using Convolutional Neural Networks (CNNs) and transfer learning. A pre-trained MobileNetV2 model is used as the base model to extract high-level features from input images, which helps improve performance and reduce training time. (Nagrath et al., 2021).

The system **pipeline** consists of the following steps:
Input & Preprocessing: The system receives face images labeled with mask or without mask. Images are resized, normalized, converted to arrays, and labels are one-hot encoded. Data augmentation (rotation, flipping, zooming, brightness) is applied to improve performance and reduce overfitting.

Feature Extraction & Classification: A pre-trained CNN model (MobileNetV2) extracts high-level features. A classification head with Pooling, Flatten, and Dense layers predicts mask or no mask.

Training & Evaluation: The model is trained using an optimization algorithm over multiple epochs. Performance is evaluated using standard metrics such as accuracy, precision, recall, and F1-score.

Output: The system outputs whether a person is wearing a mask, displayed on the image or saved in a report.

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Expected Outcomes

The project is expected to implement a pre-trained deep learning model for face mask detection and evaluate its performance on a new dataset using transfer learning. The model will classify images into mask and no mask categories, and its performance will be assessed using the same evaluation metrics reported in the original study, including accuracy, F1-score, loss curves, and confusion matrix analysis.

Different improvement techniques will be applied and compared with the baseline model to analyze their impact on performance and determine the most effective approach.

Relevant references

1- Jangra, A. (2020). Face Mask 12K Images Dataset. Kaggle.
https://www.kaggle.com/datasets/ashishjangra27/face-mask-12k-images-dataset

2 - Nagrath, P., Jain, R., Madan, A., Arora, R., Kataria, P., & Hemanth, D. J. (2021). SSDMNV2: A real-time CNN-based face mask detection system using MobileNetV2. *Sustainable Cities and Society, 66*, 102692.
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